Who, What, and Where: Plotting Ten Years of Learning @ Scale Research

Alex S. Duncan
College of Computing
Georgia Institute of Technology
Atlanta, GA, USA

Travis Tang
Georgia Institute of Technology
Atlanta, GA, USA

Yinghong Huang
Georgia Institute of Technology
Atlanta, GA, USA

Jeanette Luu
Georgia Institute of Technology
Atlanta, GA, USA

Nirali Thakkar
Georgia Institute of Technology
Atlanta, GA, USA

David A. Joyner
College of Computing
Georgia Institute of Technology
Atlanta, GA, USA

ABSTRACT

This paper examines research trends from the first 10 years of the Learning @ Scale conference across three dimensions: research context, subject matter, and the countries of the authors' affiliations. Each of the 562 papers published over the conference's history was coded across these dimensions. The results reveal a significant drop in MOOC research, a rise in research from outside the United States, and a consistent focus on computer science and STEM curricula.

CCS CONCEPTS

• **Applied computing** → **Distance learning**; • **General and reference** → **Surveys and overviews**.

KEYWORDS

MOOCs, affordable at-scale degrees, informal learning environments, CS education

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1 INTRODUCTION

The ACM Learning @ Scale conference kicked off in Atlanta in 2014. In its inaugural year, 100% of conference publications originated in North America; MOOCs were the dominant research context; and the early community was heavily populated by computer science education researchers involved in both building and teaching via this emerging class of platforms.

Since then, members of the community have had implicit beliefs about the community's development: the community has gotten more international, less MOOC-focused, and less computing-focused. This research attempts to quantify the extent of these trends in the research record. We coded all 562 papers published in the first 10 years of Learning @ Scale along three dimensions: research context (MOOCs, K-12, online degrees, in-person degrees, or informal learning environments); subject matter (computer science, STEM excluding computer science, and non-STEM); and the country associated with the affiliations of the authors. We find that, over time: the community's focus on MOOCs has diminished nearly to nothing; the community has indeed become more international (although North American institutions still dominate); and interest in various subject areas has remained relatively consistent.

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2 RELATED WORK

Four prior similar analyses have been conducted on the history of the ACM Learning @ Scale conference: Kross & Guo in 2018 [6], Kulkarni in 2019 [7], Johanes in 2019 [4], and Duncan et al. in 2023 [3]. These analyses largely focused on thematic elements of the papers in the conference, such as the topics they researched, the viewpoints they embodied, and the priorities they reflected. This analysis takes a more direct approach, examining the contexts, subjects, and origins of the research.

3 METHODOLOGY

For this research, four coders each coded all 562 total papers published in the first 10 years of Learning @ Scale. Each paper was coded for its research context and subject matter, and a list was maintained of the affiliations of each paper's authors. Informed by prior research (primarily coding performed by Duncan et al. in 2023 [3]), we first narrowed the contexts and subjects for analysis: for research context, we selected MOOCs, in-person degrees, online degrees, informal learning environments; for subject matter, we selected computing, non-computing STEM, and non-STEM. Other delineations are possible, but we found it useful to avoid too many categories with too few examples to be meaningfully summarized.

Each of the four coders coded all 562 papers based for research context and subject matter, applying only one context code (a suggestion posed by Duncan et al. [3]) to each paper. Nonetheless, a number of papers legitimately *were* situated within multiple contexts. To allow for more meaningful data analysis, rather than applying multiple codes, these papers were coded as "Multiple".

The same process used to code research context was also used to code academic subject matter. Having four people individually

code the context and subject of each paper allowed us to assess interrater reliability and ensure these codes represented some shared definition rather than an individual's unique mental model.

Interrater reliability was calculated for each combination of two coders within the group of four coders (six combinations in total) across all 562 papers using Cohen's Kappa [2]. We used Cohen's Kappa instead of Fleiss's Kappa (used to determine overall interrater reliability for more than two raters) to help us examine if any one of the four coders was not aligned with the rest of the group; we could then exclude that coder's data to increase confidence in the reliability of the data from the remaining three coders. This applied to Coder 1: their Cohen's Kappas with Coders 2, 3, and 4 were all below 0.310, while the Cohen's Kappas among Coders 2, 3, and 4 were between 0.440 and 0.850, indicating moderate to high agreement. We thus excluded Coder 1's codes when determining the definitive codes for each paper. Final codes for continued analysis for each paper were then determined by the majority of codes applied to a paper by Coders 2, 3, and 4. In instances where the three coders all selected different codes (10 such instances for research context, or 1.8% of papers; 71 such instances for subject matter, or 12.6% of papers), the "Multiple" code was assigned to the paper.

4 RESULTS

For readability, we separately report the results of the research context and subject matter analyses along with their discussions.

4.1 Research Context

Over half of the papers (52%) were coded as "Other", and 36% of the papers focused on MOOC research; the prevalence of "Other" is discussed in the Discussion section. Together, these two context categories accounted for the vast majority of papers, with only 12% of papers focused on online degrees (4%), K-12 research (3%), informal learning environments (2%), in-person degrees (1%), or multiple different contexts (2%).

A breakdown of the research contexts of Learning @ Scale papers over time is shown in Figure 1. While the percentages for most of the contexts have stayed fairly consistent over time, those for the two most frequently occurring contexts—MOOCs and "Other" contexts—show a pronounced change. In 2014—the first year of Learning @ Scale—research conducted in the context of MOOCs accounted for 78.6% of papers, with the remaining papers split evenly between informal learning environments, multiple contexts, and "Other" contexts. MOOC papers experienced a drop in number, particularly in 2018 when they dropped to only 29.4% of published papers from 59% the prior year, and in 2023, only 3.4% of papers were focused on MOOCs. Conversely, the proportion of research conducted in "Other" contexts increased over time, reaching its maximum of 87.9% of papers published in 2023.

4.1.1 Discussion. The majority of Learning @ Scale papers were coded with a context of "Other". Further investigation of these papers revealed topics such as intelligent tutoring systems, student modeling, academic integrity, and crowdsourcing—topics that are generally relatively context-agnostic. This highlights the dual nature of Learning @ Scale research from a context perspective, and the nearly even split between research conducted in a specific context and research that seems context-agnostic emphasizes the

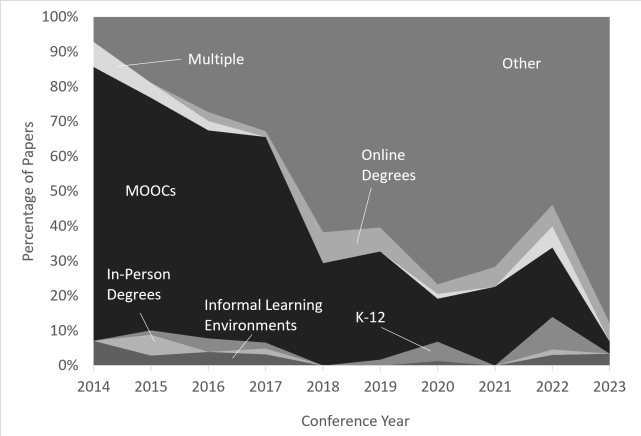

Figure 1: Context of Learning @ Scale papers over time.

interest in and importance of both types of research. Additionally, the increase in frequency over the last 10 years of papers coded with an "Other" context suggests an increasing focus on tools and methods that are applicable in a variety of learning contexts. One example is AI and ML applications in education: 30% of the papers coded with an "Other" context were also categorized under the topic of AI and ML in education.

The decrease in the frequency of MOOC papers over time is not surprising, as it tracks with the shift in popular opinion about the value and longevity of MOOCs, from The New York Times declaring 2012 "the year of the MOOC" to the vice president of Udacity proclaiming in 2017 that "[MOOCs] are dead" [5, 8]. The most notable point is not that MOOCs have become less prevalent at the conference, but that the conference has continued to thrive even as MOOCs have become less prevalent in the world. Interestingly, however, MOOC enrollments have continued to grow, with a spike in 2020 (due in part to the pandemic) [10]. This trend suggests the existence of myriad research opportunities in the realm of MOOCs, so the decrease in MOOC papers at Learning @ Scale over time is not necessarily as expected as we might have thought, and the reasons for the decrease may be an avenue for further investigation.

The relative lack of papers related to online degrees is surprising, especially the absence of a spike during the COVID-19 pandemic. Even as recently as 2023, nearly 15% of college students were enrolled in online degree programs—certainly a minority, but not a negligible percentage [1]. The lack of K-12 research, on the other hand, is perhaps expected, since scaling up K-12 education poses unique additional challenges. Online degrees and K-12 education may nonetheless be two areas for expanded interest at Learning @ Scale going forward, especially as both dovetail cleanly with otherwise context-agnostic initiatives like automated feedback.

4.2 Academic Subject Matter

Across the conference's history, papers that did not specify the subject matter (28%) have been approximately as common as papers related to computer science (27%). 18% of papers focus on non-computer science STEM subjects, meaning that nearly half (45%) of all papers published at Learning @ Scale focus on a STEM subject,

while only 6% of papers focus on a non-STEM subject. 21% of papers were coded as having multiple subject matters.

Figure 2 shows a breakdown of the subject matter of Learning @ Scale research over time. Although there have been some fluctuations, the relative percentages of the subject areas have remained fairly consistent over time. Notably, research in non-STEM areas started strong at 28.6% of papers in the first conference, but had disappeared by 2018 and has not yet been a major presence since.

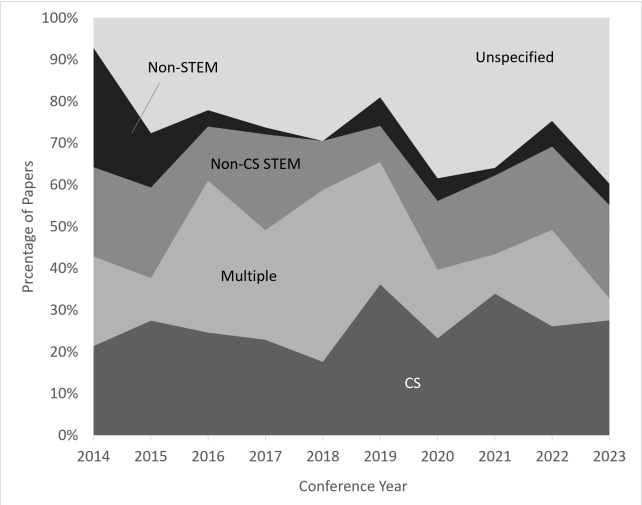

Figure 2: Subject of Learning @ Scale papers over time.

4.2.1 Discussion. Papers with unspecified subject matter have been more common at Learning @ Scale than those with specified subjects. As with papers coded with the context of "Other" (63% of papers with unspecified subjects were coded as such), this may further support the hypothesis that there is strong interest in research that is broadly applicable across many subjects, such as facilitating strong student discussion communities, predicting student success based on course interaction patterns, or detecting misconduct in commonly formatted assessments. At the same time, this may indicate room for growth in subject-specific research.

Learning @ Scale research related to computer science is nearly tied in frequency with subject-agnostic research. This is unsurprising given that computer science classes drove the early MOOC movement; additionally, computer scientists build many of the tools used to deliver at-scale learning, and CS is compatible with autograding and other approaches to teaching at scale. In contrast, non-computer science STEM subjects—and to a greater extent, non-STEM subjects—appear relatively infrequently in Learning @ Scale papers, and papers related to non-STEM subjects have even *decreased* since the conference's first couple of years. Given that the subject matter of MOOCs varies tremendously, and that at-scale online degrees have spread to many non-STEM domains in recent years [9], there is enormous opportunity to revisit non-STEM fields in future Learning @ Scale work. It is critical to examine how to deliver topics like AI literacy, civic-mindedness, and environmental sustainability at scale to meet the demands of tomorrow's citizenry.

5 AUTHORSHIP

Our authorship analysis focuses on the home country of the institution for authors over time.

5.1 Overall Distribution

41 countries and 6 continents are represented by author affiliations across all Learning @ Scale papers. The percent of papers each country appears in with regard to authorship is shown in Figure 3; a country was counted once per paper on which it appears, regardless of how many authors from that country appear on the paper. The United States appeared in 76.2% of papers and is omitted from Figure 3 to allow for clarity of visualization.

5.2 Trends Over Time

When looking at trends over time, there was enough noise in the data to hinder meaningful analysis at the country level, so we instead looked at relative representation of continents. The percent of each year's papers that each continent appears in with regard to authorship is shown in Figure 4. Note that the percentages for most years will sum to more than 100% because more than one continent can be represented on a single paper.

The percent of papers with North American authorship—which was 100% at the first Learning @ Scale conference in 2014—has since decreased to 75.9% at the 2023 conference. European authorship increased from 0% of papers in 2014 to a maximum of 37.7% in 2016 (the conference's first foray outside North America), with additional peaks in years the conference was held (or was intended to be held) in Europe (2018 and 2021). Asian authorship showed a slight increase in the past decade, although it peaked at 30.2% in 2021 and declined over the following two years. We also observed that authorships from Africa, Oceania, and South America have never risen above 8.2% of papers in a given year, and there are multiple conference years for each of these continents in which they are not represented on any papers at all.

5.3 Discussion

It is encouraging to see so many countries (and almost all continents) represented among Learning @ Scale authors, although United States authors appear on the vast majority of papers, and the other three countries in the top four are in either North America or Europe. The Association for Computing Machinery (ACM), which manages Learning @ Scale, is based in the United States, and this could be a contributing factor. Representation of United States authors has decreased over time, although Western countries continue to be the most frequently represented among paper authors. Of course, the conference has only been held in Western countries, and we noted an uptick in papers from European countries when the conference was held in Europe. Hosting the conference in Asia or Oceania may increase international participation.

To what extent does the fact that Learning @ Scale and its proceedings are all in English play a role? What other variables might influence this trend? We believe it is worth investigating questions such as these as a step towards further diversifying the representation of countries in Learning @ Scale authorship, and towards combating the implicit notion that the Western perspective of at-scale education is the only perspective.

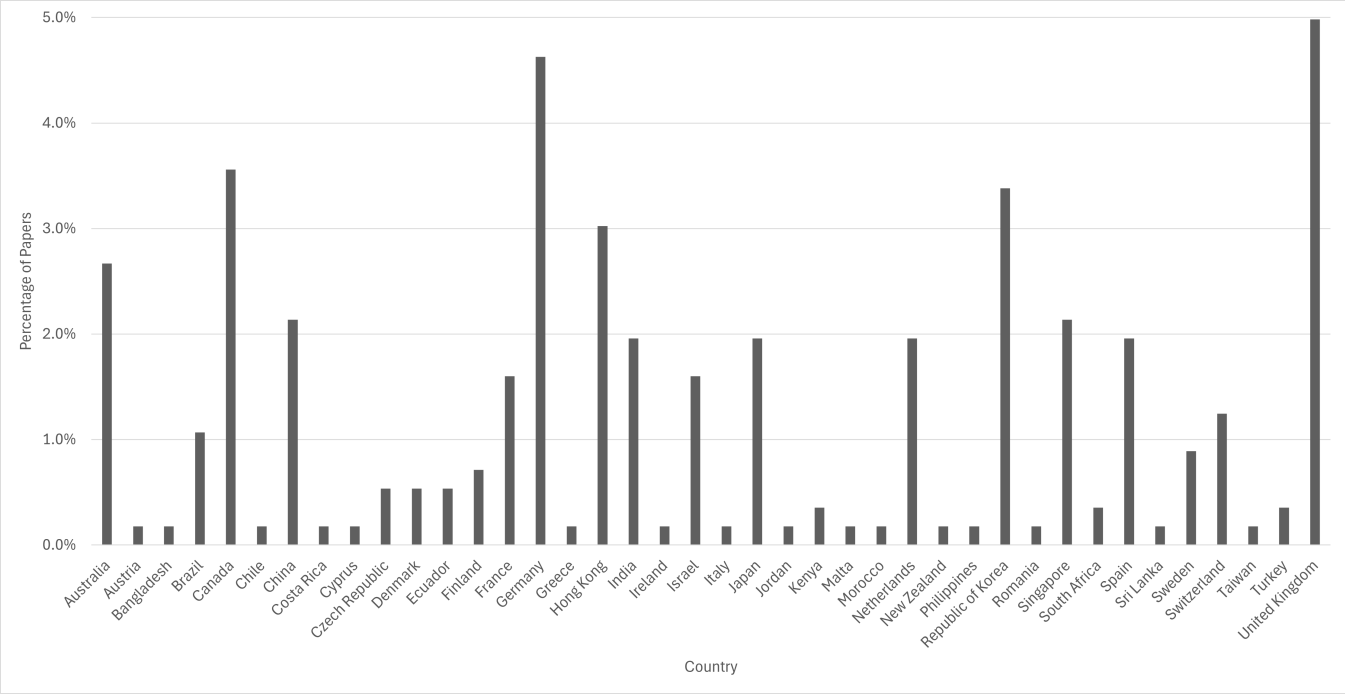

Figure 3: Percentage of Learning @ Scale papers each country appears in the authorship for. (The United States appeared in 76.2% of papers and is omitted for clarity.)

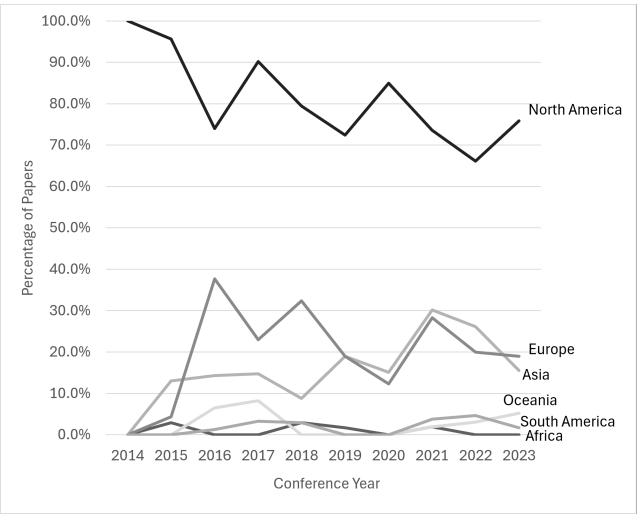

Figure 4: Percentage of Learning @ Scale papers each continent appears in the authorship for over time.

6 LIMITATIONS

Coders noted difficulties when coding for research context and subject matter for papers that seemed to belong to more than one category. We ultimately allowed coders to apply the "Multiple" code if they believed it was necessary. While our authorship analysis was comprehensive, representation on Learning @ Scale papers is dominated by a few countries, in particular the United States, and the limited data for less-represented countries is an asterisk on any conclusions we might draw about their representation. More generally, while we discussed a number of findings related to Learning @ Scale authorship, we recognize the complexities inherent in determining causality, as there may be political or cultural factors outside the scope of our research.

7 CONCLUSION

Many of the perceived shifts in the Learning @ Scale community are backed up by the data: the conference has become more international and less MOOC-focused over time. The former shift is moderate: the conference opened with 100% of publications originating in North America, and this number remained high at 75.9% in 2023. The drop in MOOC research is more pronounced, from over 75% in 2014 to under 5% in 2023. This trend has run parallel to an increase in context-agnostic initiatives such as intelligent tutoring tools that are targeted at specific subjects rather than contexts. Subject matter interest has remained relatively stable over the course of the conference's history, however. Non-STEM subjects remain underrepresented and provide a fruitful area for future research.

REFERENCES

[1] Jennifer Calonia. 2024. *Are Online Degrees Respected?* Retrieved February 15, 2024 from https://www.forbes.com/advisor/education/online-colleges/are-online-degrees-respected/

[2] Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46. https://doi.org/10.1177/001316446002000104

[3] Alex Duncan, Ana Rusch, Prerna Ravi, and David Joyner. 2023. The L@St Eight Years: A Review of Papers and Authors at Learning @ Scale. In *Proceedings of the Tenth ACM Conference on Learning @ Scale (L@S '23')*. ACM, New York, NY. https://doi.org/10.1145/3573051.3596192

[4] Petr Johanes. 2019. Start of a Science: An Epistemological Analysis of Learning at Scale. In *Proceedings of the Sixth (2019) ACM Conference on Learning@ Scale*. 1–10.

[5] Varuni Khosla. 2017. *Udacity to focus on individual student projects*. Retrieved February 15, 2024 from https://economictimes.indiatimes.com/industry/services/education/udacity-to-focus-on-individual-student-projects/articleshow/60963078.cms

[6] Sean Kross and Philip J Guo. 2018. Students, systems, and interactions: synthesizing the first four years of learning@ scale and charting the future. In *Proceedings of the Fifth Annual ACM Conference on Learning at Scale*. 1–10.

[7] Chinmay Kulkarni. 2019. Design perspectives of learning at scale: scaling efficiency and empowerment. In *Proceedings of the Sixth (2019) ACM Conference on Learning@ Scale*. 1–11.

[8] Laura Pappano. 2012. *The Year of the MOOC*. Retrieved February 15, 2024 from https://www.nytimes.com/2012/11/04/education/edlife/massive-open-online-courses-are-multiplying-at-a-rapid-pace.html

[9] David S Park, Robert W Schmidt, Charankumar Akiri, Stephanie Kwak, and David A Joyner. 2020. Affordable Degrees at Scale: New Phenomenon or New Hype?. In *Proceedings of the Seventh ACM Conference on Learning@ Scale*. 25–35.

[10] Dhawal Shah. 2021. *By The Number: MOOCs in 2021*. Retrieved February 15, 2024 from https://www.classcentral.com/report/mooc-stats-2021/